

**The “MASTER CODE” of Biology:
A Basic Research Discovery Unifying the 3 Worlds of DNA, RNA
and Amino Acids from Isotopes to Genomes**

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Abstract

In This article we present a strong basic research: the “*Master Code of Biology*” discovery: First, the discovery that we named "formula for Life": this simple formula that will be presented here unify all the components of life structuring them in a kind of “periodic table of Biology”: bioatoms CONHSP , nucleotides UTCAG, amino acids, DNA strands, RNA strands, proteins and genes, chromosomes and genomes. Then, this discovery open strong perspectives in exobiology, proposing a specific Life emerging constraint on isotopes proportions tuning and balancing.

Secondly, we introduce the “Master Code of Biology”: it is a kind of digital language that would provide a common alphabet to the 3 fundamental languages of Genetics, Biology and Genomics that are DNA, RNA and proteins: the output is that, above the 3 languages of Biology - DNA, RNA and amino acid -, there is a universal common code which unifies, connects and contains all these three languages: we named "Master Code of Biology", it provides a great Unification between Genomics (DNA) and Proteomics (amino acids) Master Code images which appear highly correlated, while RNA images curiously appears like a neutral or zero-like code.

Thirdly, functionality of this basic research discovery is evidenced by two kinds of examples: -Analysing textures of above Genomics and Proteomics patterned curves provides emergence of "binary codes" and discrete "waveforms" which could predict the well known Karyotypes white/grey/black bands overlapping and characterizing the human chromosomes. -Computing the Master Code of Biology to SNP genomic areas, we show that SNPs are more functional by their location within the genome than by their local values T/C/A/G nucleotides changes.

Key Words: Universal genetic code, Human genome, genomics, DNA decoding, Evolution, SNP, Origins of Life, Exobiology, Fractal genome, Golden ratio, Fibonacci numbers.

1. Introduction

We are studying possible numerical structures in DNA and genomes since about 25 years (Perez 1991; Marcet 1992; Perez 1997). Then, we present now strong results based on a discovery done about in 1997 then explored and developed over fifteen years; the most comprehensive description to date in the book (Perez 2009, chapter 20 and annexe1) and in (Perez 2011).

Initially our main goal was understanding the heterogeneity of the 2 languages DNA and amino acids linked by the heterogeneous universal genetic code table. A number of researchers are working in this difficult strong field (Petoukhov 2011; Negadi Tidjani 2014; Rakocovic 2011).

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How to discover a possible law unifying these two worlds as different as DNA codons and amino acids? We found three candidate ways linking and possibly unifying these two worlds:
 -first atomic mass provides a common measure to codons and amino acids which are build with common bio atoms.

-then secondly, we thought about a kind of law of digital projection of the atomic masses which could provide periodicities and regularities. Indeed, a phenomenon which appears in a heterogeneous tridimensional space can emerge regular and organized when the same data space is analysed in two dimensions.

-finally we had a hunch that the universal variables like *Pi* and the golden ratio *Phi* could, maybe, play a role in this projection formula ...

It is the synthesis of these three intuitive tracks leading to the discovery of this formula unifying all the molecules of life called “formula for life”.

Then this 1997 discovery was improved in multiples areas of biology, genetics and genomics. These results were only partially published¹ in (Perez 2009; Perez 2010; Perez 2011; Pellionisz et al. 2012; Perez 2013; and in the drafts (Perez 2001; Perez 2008). This main discovery will be more fully described in a series of articles in preparation: particularly (present article §3.1. to §3.4., and three or four articles in preparation: Perez 2015abcd).

2. Materials and Methods

2.1. Genome analysed:

We analyzed the whole human genome in its most recent release (Baltimore D., 2001), [6].

2.2. Bravais Pearson Correlation Method :

All correlations in this paper run the “Bravais-Pearson” linear correlation algorithm.

Where:

If X and Y are two vectors of elementary items x_i and y_i to correlate, then :

$$r = \frac{\text{cov}(X,Y)}{\sqrt{V(X)V(Y)}} \quad \text{cov}(X,Y) = \frac{1}{n} \sum x_i y_i - \bar{x} \cdot \bar{y}$$

And $V(X)$ and $V(Y)$ are the « variance » then $\text{cov}(X,Y)$ is the « covariance ».

3. Results and Discussion

Now, we present the Master Code Discovery detailing it in four steps: “*formula for Life*”, “*Biology periodic table*”, and “*Biology master code*”, “*discrete WAVEFORMS and logic BIOBITS*” discoveries... A summarized overview of these 4 breakthrough are presented in following §3.1, §3.2, §3.3, and §3.4.

3.1. An overview on “Formula for Life” discovery:

A quick presentation of the "Formula for Life":

¹ Anteriority considerations : we never published detailed methods for calculating the Master Code of Biology, publishing only results and perspectives of applications. This article is the first to reveal these methods permitting to other researchers to reproduce our results.

In (Perez 2009; Perez 2010; Perez 2011; Pellionisz et al. 2012) we started to introduce this law we call "formula for Life" that unifies all of the components of living since the bioatoms CONHSP atoms and their various isotopes, to genes, RNA, DNA, amino acids, chromosomes and whole genomes ... this law is the result of a simple non-linear projection formula of the atomic masses. Then, the result of this projection is organized in a linear scale of integer numbers based codes (i.e. -2 -1 0 1 2 3...) coding multiples Pi/10 regular values. These codes are called « Pi-masses ».

Computing the “formula for Life” associated with any atomic mass of Life components:

For atomic mass of any biological compound, we operate the "projection" of the atomic mass numerical value using the following operator:

$$\text{Proj (m)} = [1 - [4\pi\sqrt{\varphi}\varphi\varphi^2]] m$$

2008 : The EQUATION of LIFE... by jean-claude Perez

with: $\sqrt{\varphi}=1/\sqrt{\Phi}$ *Phi is the GOLDEN RATIO Φ*
 $\varphi = 1 / \Phi$
 $\varphi^2 = 1/\Phi^2$

The result is a real number which we retain only the residues (decimal remainder) ...

Detailing the "PPI(mass)" projection:

Consider any atomic mass « m », which may be that of a bio-atom, of a nucleotide, a codon of an amino acid or other genetic compound based on bio-atoms or even, any atoms (Mendeleïv Table, (Perez 2014)) .

This process will work especially on the average masses (see Table1). But it may also be applied to a particular isotope or any derivative of specific atomic mass proportions of the various isotopes.

If we set FI = 1/Phi the golden ratio,

$$\text{Proj PPI (m)} = [1 - (4 \times \text{PI} \times (\text{FI} * .5) \times \text{FI} \times (\text{FI} * 2)))] m$$

Consider the constant P such that $P = 0.742340663 = (4 \times (\text{FI} * .5) \times \text{FI} \times (\text{FI} * 2))$ (we also rounded to the ninth decimal P).

Consider the "v" value, where v is always a negative or zero real number.

Consider the function $\text{Remainder}(v) = \text{Abs}(\text{whole number just after whole number } v) - \text{Abs}(v)$

Where Abs (x) is the absolute value of x, and « remainder » or « residue » the decimal remainder of the numerical projection.

For example: $\text{remainder}(-27.85) = 28 - 27.85 = 0.15$

We then defined PPI (m) such that: $\text{PPI (m)} = \text{remainder} [(1-P.\text{PI}) m]$

Note that (1-P.PI) is always negative because m is always positive, and (1-P.PI) is always negative.

As an example, consider the amino acid GLY:

We defined the average mass of GLY as: $GLY = 75.067542$

Then: $(1-P.PI) GLY = -99.99987286$

Thus, $PPI (GLY) = \text{remainder} [(1-P.PI) GLY] = 0.0001271351803$

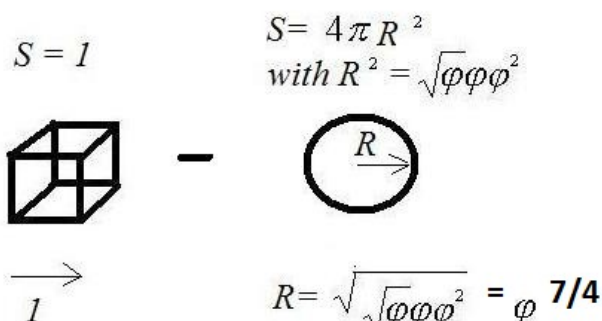
Or, finally, $PPI (GLY) = 0.0001271351803$

Although no longer considered the decimal part, we note that, if we were interested in the set $(1-P.PI) GLY = -99.99987286$, this value is substantially equal to $100 = 10 * 2 \dots$ which is not a number "any" ...

What is the geometric reality of this projection? As Figure 1 summarizes below, everything happens as if the atomic mass was "filtered" through the competitive interference of two projections:

one through a cube of side = 1 and the second-through of a sphere of radius = $FI * 7/4$.

"Is there an EQUATION for LIFE?"



Differentiating 2 projections of the atomic mass m
 - Projection on a CUBIC 3-D surface.
 - Projection on a SPHERIC 3-D surface.

$$\text{Proj}(m) = [1 - [4\pi \sqrt{\phi\phi\phi^2}]] m = [1 - [4\pi \phi^{7/2}]] m$$

with: $\sqrt{\phi} = 1/\sqrt{\Phi}$ Φ is the GOLDEN RATIO Φ
 $\phi = 1/\Phi$
 $\phi^2 = 1/\Phi^2$

This universal formula process a differentiation or interference between both atomic mass "m" projections through a CUBE 2D surface of size 1x1x1 and a SPHERE of size $R = \phi^{7/4}$

The result is a common UNIFIED scale common to ALL organic compounds (DNA, RNA amino acids)...

Details in the book of J.C Perez CODEX BIOGENESIS:
<https://sites.google.com/site/codexbiogenesis/>

Figure 1 – "J.C. Perez's Formula for Life" non linear projection highlights.

Table 1 below illustrates the calculation of Pi-mass projections for some representative biological components: bioatoms, nucleotides, DNA codons, amino acids.

Table1: A set of Pi-mass projections for some main Life compounds.

MOLECULE OR BIO-ATOM	AVERAGE Atomic Mass	Projection PPI(m)	PI-MASS NPI(m) N.PI/10	Error EPI(m,N)
C(12) Carbon (isotope 12)	12.000000	0.01441631887	0 PI/10 (0°)	0.01441631887
C (Carbon average mass)	12.0111	0.0003703460363	0 PI/10 (0°)	0.0003703460363

G (G nucleotide)	150.120453	0.01974469326	0 PI/10 (0°)	0.01974469326
Codon TCA	369.324471	0.01106361166	0 PI/10 (0°)	0.01106361166
Codon UCA	355.297477	0.6968708101	-1 PI/10 (-18°)	0.0110300755
Codon AGT (TCA complement)	409.349065	0.6930222208	-1 PI/10 (-18°)	0.0071814862
DNA double strand : TCA+AGT	778.673536	0.7040858325	-1 PI/10 (-18°)	0.0182450978
PRO (Proline amino acid)	115.13263	0.6281423922	+2 PI/10 (+36°)	0.0001761385
LYS (Lysine amino acid)	146.190212	0.2553443926	+4 PI/10 (+72°)	0.0012926688
CONH Peptidic link	43.025224	0.6847234457	-1 PI/10 (-18°)	0.0011172889

3.2. An overview on “the Periodic table of Biology” discovery:

In Figure 2 below, we introduced - to 162 representing biological components - first, the real atomic masses and, on the other hand, their corresponding Pi-mass weights projections. This figure then reveals the "periodic" nature of these regular patterned projections. We note that the 2 increasing areas to the right of the graph correspond to the respective curves of the 64 DNA and RNA codons atomic weight.

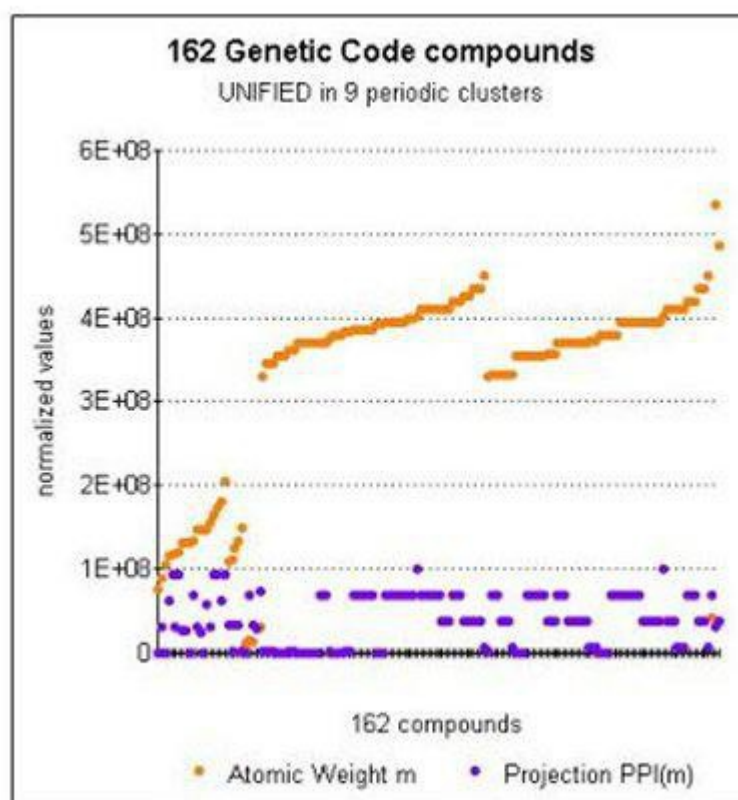


Figure 2 – “The Periodic Table of Biology”: evidence of a regular projection scale based on Pi/10 units.

Table2 shows that we will allow ourselves to call ***“periodic table of Biology”***: It represents successive regular periods based on Pi / 10 tokens. It shows all the essential components of Biology and Genetics: bioatoms, DNA and RNA nucleotides, amino acids, codons and codon pairs from DNA and RNA molecules.

Is it legitimate to speak of "*periodic table of Biology*" by analogy with the universal "periodic table" of Mendeleev? We say "yes" for the following two reasons:

Firstly the regular pitch multiples of $\text{Pi} / 10$ définie, in our view, the horizontal dimension of the table (horizontal column size).

On the other hand, to complete the Mendeleïev's table analogy, the vertical scale, the periods vertical size in the table of Mendeleïev, could be formed by the levels of complexity of the molecules.

Thus, the period 1 would be that one of the isotopes, the period 2 the average atomic masses of atoms, the period 3 that of simple molecules, the period 4 that complex composite molecules ... etc ...

The Table2 below suggests an example of 4 periods (lines) on the $\text{Pi-mass} = 0$ column. Finally, to complete the analogy, we will show in (Perez 2015) how we can associate a kind of simple algebra and allowing combinatorial, analytically calculate Pi-mass directly from the 6 basic CONHSP Pi-mass bioatoms.

An interesting research track of this "Period dimension" is that of a possible link between these molecules with same Pi-masses and their possible tri-dimensional shape: in fact, two contiguous molecules with same Pi-mass could control the geometric orientation of successive spatial planes organization. Thus, the molecules gradually increase assembly complexity, whether both cases of same or different single molecules or very different complexities molecules, for example interaction between a DNA codon and a water molecule (see Table2).

Table2: "The Periodic Table of Biology", suggesting a biomolecules complexity vertical Period dimension.

Périod	Nature	Example	Atomic Mass	Pi-mass	Error
1	Isotopes	Carbon C12	12.00000	0 Pi/10	0.014416327
2	Average atomic mass	Carbon atom	12.01110	0 Pi/10	0.000370338
3	Simple molecules	H2O	18.0153240	0 Pi/10	0.001210900
4	Complex molecules	Codon TCA	369.324471	0 Pi/10	0.01106361166

Table3: "The Periodic Table of Biology" structuring ALL compounds of Life in a regular $\text{Pi}/10$ units scale.

	-3Pi/10	-2Pi/10	-1Pi/10	0Pi/10	+1Pi/10	+2Pi/10	+3Pi/10	+4Pi/10	+5 and +7Pi/10
Bioatoms	P(-4pi/10)		H O	C	N			S	
Nucleotides				U G I	T C A				
Annex Molecules		Phosphate /sugar RNA	CONH	H2O	CH2 Phosphat e/sugar DNA				
Amino Acids			Asp	Asn Glu Gly Ser	Ala Gln His Thr	Pro Tyr Cys (+2)	Arg Phe Trp Val	Ile Leu Lys Met (+4)	Cys (+5) Met (+7)
DNA Codons	ggg	gtg gcg gag tgg cgg agg ggt ggc gga	ttg ctg atg gtt gtc gta tgc ccg acg gct gcc gca	ttt ttc tta ctt ctc cta att atc ata tct tcc tca					

			tag cag aag gat gac gaa tgt tgc tga cgt cgc cga agt agc aga	cct ccc cca act acc aca tat tac taa cat cac caa aat aac aaa					
RNA Codons	uuu uug guu gug ugu ugg ggu ggg	uuc uua cuu cug auu aug guc gua ucu ucg gcu gcg uau uag gau gag ugc uga cgu cgg agu agg ggc gga	cuc cua auc aua ucc uca ccu ccg acu acg gcc gca uac uaa cau cag aau aag gac gaa cgc cga agc aga	ccc cca acc aca cac caa aac aaa					

A strange observation opening up enormous opportunities in astrobiology:

Table4 and Figure3 below shows a very curious fact: *the formula Pi-mass projection seems optimal only for the atomic masses of average atomic weights of Life basic bioatoms C O N H*. Instead of tiny perturbations on these atomic masses as well as atomic masses of the individual isotopes (example O16) of each of these atoms "destroy" the optimality and fine-tuning of these projections then, also, consequently all described in §3.3. and §3.4.. Example here (Table 4) for the Pi-mass projection of Oxygen isotopes and % average weighted atom mass. As shown in Figure3, isotopes of oxygen lightest and heaviest O16 O18 both produce an error on the projections Pi-mass much higher than that of the average atomic mass of that atom of oxygen consisting of: 99,757% + 0.04% O16 O17 + 0.2% O18.

Then, to summarize : we discovered a formula unifying ALL biological components from C O N H S P bioatoms to DNA, RNA, amino acids then genes proteins chromosomes and genomes. But this perfect balance is optimal only when atom atomic mass is the % mix from our Earth atmosphere: i.e %C12 C13 C14 etc... This fine tuning is destroyed by example when all isotopes are C12 only... or also in case of environmental disorders. For example, the proximity of a large chemical plant alter the relative proportions of isotopes of carbon or oxygen. This will result in an imbalance of the Pi-masses and consequently embrittlement DNA (Perez 2015) capable of inducing cancers by some chromosomes translocations. What is the reason for that perfect selectivity of our « formula for Life » would in fact Operational only when the relative proportions of bioisotopes are exactly those that we measure in our Earth's atmosphere inherited from millions of years of evolution?

The main question now is : Is this perfect isotopes proportions balance the result of Earth and atmosphere evolution according to the thesis of Dr James Lovelock (Lovelock 2014) ? In (Perez 2009) and (Perez 2015) I sketch an alternative hypothesis: the multiple isotopes of the same atom would perhaps only partial views of a multi-atom isotopes. By analogy, one can think of the white light that is in fact the synthesis of multiple components of basic spectral colors. Here, the phenomenon is similar but opposite kind ... This again would mean that the true atomic weight is the average atomic mass of the atom, all isotopes together, while individual atomic masses of each isotope would be only "partial views" of this global and

multi-isotope atom. Reasons of this possible multi-isotopes perception could be of quantum nature...

My question now is :

« *Is this the result of James Lovelock GAIA theory?*

Or

Perhaps isotopes does not exists but are quantum views of a meta multiple isotopes quantum atom structure? »

A strong consequence for EXOBIOLOGY: This condition of atmosphere like %mix balancing of isotopes prerequisite within C O N H bioatoms is a necessary condition but not sufficient for earth like Life on other planets...

More specifically: "The emergence of life requires the existence of specific proportions of different isotopes of each of the 4 bioatoms C (carbon) O (oxygen) N (nitrogen) H (hydrogen) such that the balance between these proportions (C12 / C13, O16 / O17 / O18, N14 / N15 and H1 / D) is strictly identical to that observed in the Earth's atmosphere. "

Table4: Example of Pi-mass projection fine tuned selectivity for Oxygen average mass vs. individual isotopes.

atom	Isotope	Relative atomic mass	% Isotopic composition	Pi projection residue and Pi mass value	Pi mass (N x Pi/10)	Error
Oxygen	average	15.9994(3)	–	0.686647751 0.685840735	-1	0.000807016
8O	16	15.994 914 619 56(16)	0.997 57(16)	0.692662834 0.685840735	-1	0.006822099
	17	16.999 131 70(12)	0.000 38(1)	0.354913152 0.371681469	-2	0.016768318
	18	17.999 161 0(7)	0.002 05(14)	0.022742056 0.000000000	0	0.022742056

jcPerez's ORIGINofLIFE newCONSTRAINTS
Average Mass OPTIMALITY vs Monoisotopes

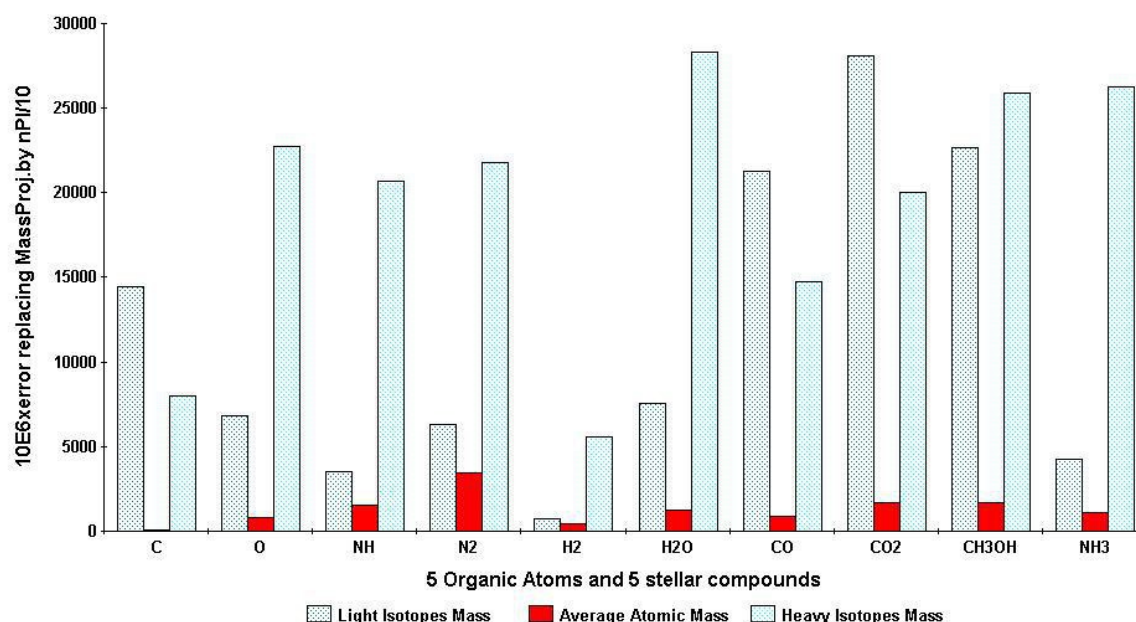


Figure 3– Fine tuned selectivity of the “Formula for Life” comparing average atomic mass with light and heavy single isotopes masses.

Finally, in (Perez 2009) in Chapter 20, we show how this superiority Pi-mass based average atomic mass compared to Pi-mass of basic isotopes, can be extended to thousands of organic molecules referenced in the famous Beilstein² dictionary: we study the primitive molecules of the astrobiology to complex components of living also passing by thousands of molecules of organic chemistry.

3.3. An overview on "the Biology Master Code" Unification of DNA, RNA and Amino acids:

It may seem surprising that such a fine tuned process like biology of Life request the use of three languages as diverse and heterogeneous as are the DNA with its alphabet of four bases TCAG, RNA with its alphabet of four bases UCAG, and proteins with their language of 20 amino acids. Obviously, the main discoveries in biology were those who managed to find the respective areas and "bridges" between these three languages. However, any researcher somewhat "aesthete" will find the table of the universal genetic code rather "ad hoc" and heterogeneous.

Now that we have described above, it is a kind of digital language that would provide a common alphabet (Pi-masses scale) to the 3 fundamental languages of Genetics, Biology and Genomics that are DNA, RNA and proteins.

The construction method of the Master Code will be fully described in (Perez 2015) and will be summarized below. It will highlight a significant discovery we summarize as follows: ***"Above the 3 languages of Biology - DNA, RNA and amino acid -, there is a universal common code unifies, connects and contains all these three languages: we named "Master Code of Biology".***

Here is a brief description of the process of computing the Master Code:

The coding step:

Firstly, we apply it to any DNA sequence encoding a gene or any uncoding sequence (junk DNA), so it may be either a gene, a contig of DNA, or an entire chromosome or genome. In this sequence, we consider always double-stranded DNA as we explore the following three codon reading frames and following the two possible directions of strand reading (3' ==> 5' or 5' ==> 3'). The base unit will always be the triplet codon consists of three bases. As the table5 the above, we calculate the Pi-mass related to double stranded DNA bases, double stranded RNA bases, or what seem strangest of two amino acids that correspond to the two complementary triplet codons of double-stranded DNA or RNA, and this systematically even when it is not a gene, but any DNA sequence (junk DNA).

A simple example: the starting region of Prion gene:

DNA image coding:

ATG CTG GTT CTC TTT...

-1 -1 -1 0 0...

complément:

TAC GAC CAA GAG AAA...

0 -1 0 -2 0...

RNA image coding:

AUG CUG CUU CUC UUU...

-2 -2 -3 -1 -3...

complement:

UAC GAC GAA GAG AAA...

-1 -1 0 -2 0...

Proteomics image coding:

²Beilstein journal or organic chemistry: <http://bjoc.beilstein-journals.org/>

MET LEU VAL LEU PHE

4 3 3 4 3...

complement:

TYR ASP GLN GLU LYS

2 -1 1 0 4...

Pi-masses corresponding to two strands are then added for each triplet:

Double strand DNA image coding: -1 -2 -1 -2 0...

Double strand RNA image coding: -3 -3 -3 -3 -3...

Double strand Proteomics image coding: 6 2 4 4 7...

This produces three digital vectors relating to each of the 3 DNA, RNA, and proteomics coded images.

There is already an absolutely remarkable result, as symbolized in Figure 6 below:

The coded image of any RNA sequence is "flat" (always=-3) and monotonous, and can be likened to the "zero" or a sort of "neutral element of Biology"!

We will focus now – exclusively - on the DNA code (genomics) and amino acids code (proteomics).

The globalization and integration step:

To these two numeric vectors we apply a simple globalization or integration linear operator. It will "spread" the code for each position triplet short, medium, long distance, producing a sort of impact or resonance of that position on the most distant positions (and reciprocally by feedback), and for each position. This gives a new digital image that we retain not the values but the rankings (by sorting them).

We run this process for each codon triplet position, for each of the 3 codon reading frame and for the 2 sequence reading directions (3' ==> 5' and 5' ==> 3').

For example, to summarize this method: on starting area of the GENOMICS (DNA) code of Prion above, the "radiation" of triplet codon number 1 would propagate well:

-1 -2 -1 -2 0... ==>

-1 -3 -4 -6 -6... then, we cumulate these values: -20

So we made a gradual accumulation of values.

The same operation from the codon number 2 produce:

-1 -2 -1 -2 0... ==>

-2 -3 -5 -5... then, we cumulate these values: -15³

etc...

Similarly, the same process on starting area of the PROTEOMICS code of Prion above, the "radiation" of triplet codon number 1 would propagate well:

6 2 4 4 7... ==>

6 8 12 16 23... then, we cumulate these values: 65

So we made a gradual accumulation of values.

The same operation from the codon number 2 produce:

6 2 4 4 7... ==>

2 6 10 17... then, we cumulate these values: 35⁴

etc...

Then, finally, after computing by this method these "global signatures" for each codon position at genomics and proteomics levels, we sort each genomic and proteomic vector to

3 In fact, in order to facilitate replication of these results, we must clarify that the sequence is looped back on itself, which is true for most bacterial genomes, but arbitrary in the general case. So here, if the Prion sequence consisted only of these five codons, the result would be after looping = -15 + -1 = -16.

4 In fact, in order to facilitate replication of these results, we must clarify that the sequence is looped back on itself, which is true for most bacterial genomes, but arbitrary in the general case. So here, if the Prion sequence consisted only of these five codons, the result would be after looping = 35 + 6 = 41.

obtain the codon positions ranking: example: as illustrated bellow, the Genomics ranking patterned signature is 2 1 4 3 5 for this Prion starting 5 codons mini subset sequence of 5 codons positions (arbitrary values). Then, to summarize the Master Code computing method on this 5 codon positions starting Prion protein sequence:

Genomics signature:

codon 1:	!
Basic codes:	-1 -2 -1 -2 0
Potentials (with circular closure):	-1 -3 -4 -6 -6
circular complement	0
cumulates:	-20
cumul	-20

codon 2:	!
Basic codes:	-1 -2 -1 -2 0
Potentials (with circular closure):	-2 -3 -5 -5
circular complement	-6
cumulates:	-21

codon 3:	!
Basic codes:	-1 -2 -1 -2 0
Potentials (with circular closure):	-1 -3 -3
circular complement	-4 -6
cumulates:	-17

codon 4:	!
Basic codes:	-1 -2 -1 -2 0
Potentials (with circular closure):	-2 -2
circular complement	-3 -5 -6
cumulates:	-18

codon 5:	!
Basic codes:	-1 -2 -1 -2 0
Potentials (with circular closure):	0
circular complement	-1 -3 -4 -6
cumulates:	-14

final rankings:

codon positions:	1	2	3	4	5
Potentials:	-20	-21	-17	-18	-14
rankings:	2	1	4	3	5

Then similar computing for Proteomics...

codon 1:	!
Basic codes:	6 2 4 4 7
Potentials (with circular closure):	6 8 12 16 23
circular complement	0
cumulates:	65

codon 2:	!
Basic codes:	6 2 4 4 7

Potentials (with circular closure):	2 6 10 17
circular complement	23
cumulates:	58

codon 3:	!
Basic codes:	6 2 4 4 7
Potentials (with circular closure):	4 8 15
circular complement	21 23
cumulates:	71

codon 4:	!
Basic codes:	6 2 4 4 7
Potentials (with circular closure):	4 11
circular complement	17 19 23
cumulates:	74

codon 5:	!
Basic codes:	6 2 4 4 7
Potentials (with circular closure):	7
circular complement	13 15 19 23
cumulates:	77

final rankings:	
codon positions:	1 2 3 4 5
Potentials:	65 58 71 74 77
rankings:	2 1 3 4 5

Then finally:

codon position: 1 2 3 4 5

Genomics vector: 2 1 4 3 5

Proteomics vector: 2 1 3 4 5

To complete, the same work must be also operate on each codon reading frame...

The great Unification between Genomics and Proteomics Master Code images:

When applying the process described above in any sequence – gene coding, DNA contig, junk-DNA, whole chromosome or genome -, it appears one second surprise just as stunning as that of RNA neutral element we find that - for one of the three reading frames of the codons given - the Genomics patterned signature and the Proteomics patterned signature are highly correlated.

Contrary to the 3 genomics signatures which are correlated in all cases, the proteomics signatures are correlated with genomics signatures only for one codon reading frame, and generally in dissonance for the two remaining codon reading frames. Also, there are perfect local areas matchings focusing on functional sites of proteins, hot-spots, chromosomes breaking points etc... Both figures 4 and 5 below summarize this universal breakthrough for the general case and for 3 representative cases: Prion protein, a whole chromosome of Malaria disease, and a complete HIV1 genome. It is important to note the universal character of this coupling genomics / proteomics: for example, for some 3 billion base pairs of the whole human genome, we have verified this law across the entire genome, for all its chromosomes and in all its regions with a global correlation of about 99%. In this global correlation, specific

codon positions were a perfect match. This is remarkable regions correspond to biologically Functional areas: hot-spots, the active sites of proteins, breakpoints and chromosome fragility regions (ie Fragile X genetic disease), etc ...

The "MASTER CODE": unifying Dna,Rna,AA Mirror-like Dna/A.A and RNA fixed point

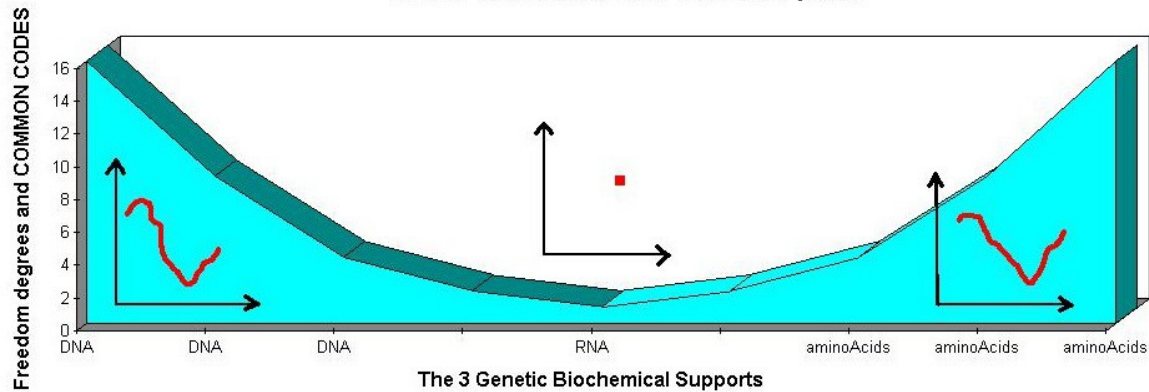


Figure 4 – “J.C. Perez's great unification” shows an equivalence of both Genomics (DNA) and Proteomics (amino acids) signature while RNA signature is a neutral area like a “zero”.

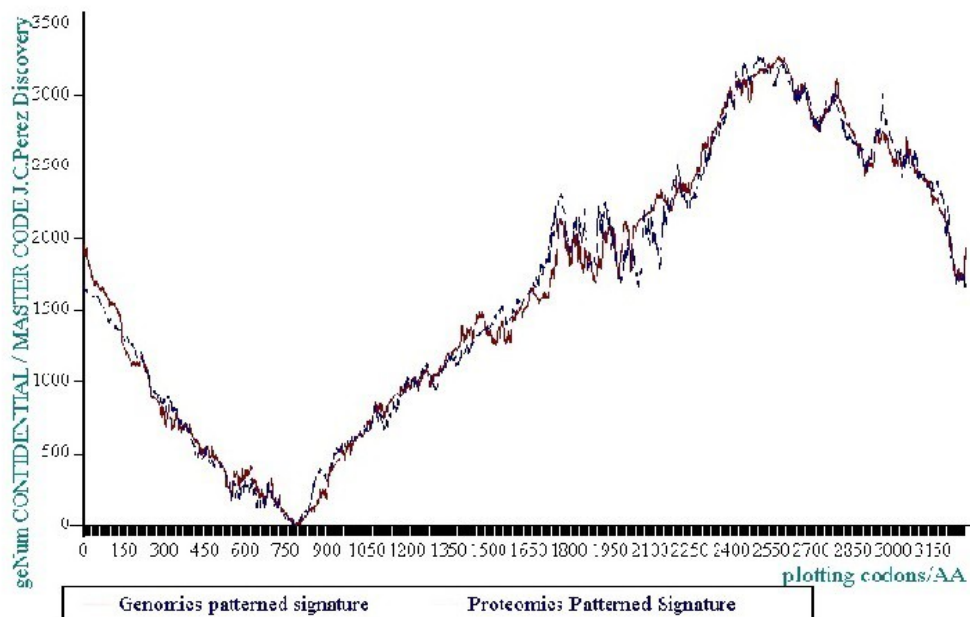
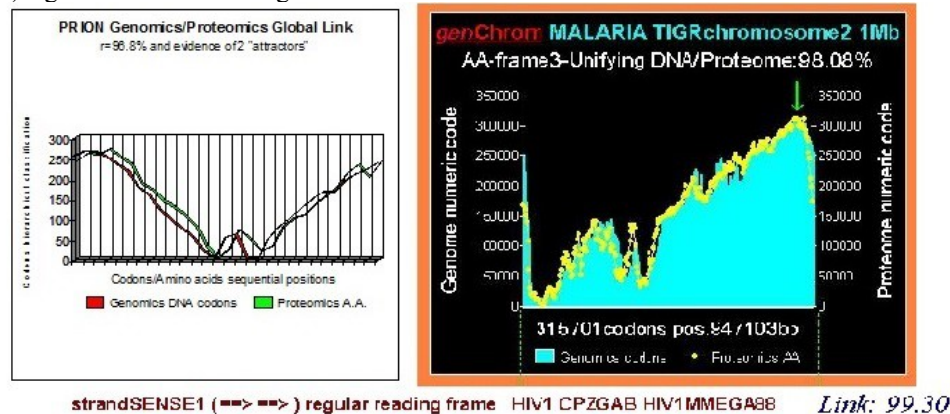


Figure 5 – A typical correlation between Genomics and Proteomics signatures related to the Prion protein, the whole Malaria chromosome 2, and the whole HIV1 genome.

3.4. Towards discrete WAVEFORMS and logic BIOBITS overlapping whole chromosomes and genomes:

Here we will analyze the "texture", that is to say, the "roughness" of genomic and proteomic signatures provided by the Master Code. For this, we need only to analyze the slopes or mathematical differentiations from these patterned curves: slopes and gradients - in the sense of Leibnitz - of order 1, of order 2, of order n.

The curves of the Master Code are discontinuous (each point represents a position of triplet codon), If we note $M(i)$ the Master Code function as defined in §4.4, then we agree that:

slope = 1 = "growing" i.e "increase" if $M(i+1) > M(i)$

and slope = 0 = "decreasing" i.e "decrease" if $M(i+1) < M(i)$.

BIOBITS: The emergence of a "binary language" from the PROTEOMICS Master Code of any DNA sequence:

A detailed analysis of the "texture" of Genomics and Proteomics curves will now be a strange phenomenon: as shown in Figure 6 below, a curious roughness "sawtooth" usually characterizes these images. This amounts to somehow search for the "derivative of order 1", that is to say the "slope" between two successive points. It becomes apparent that these slopes are mostly in the same direction: always growing or always decreasing. Here is a small example of a sequence of 312 bases where genomics (blue) and proteomics (green) signature (amino acids) are studied. Note the beauty of these mathematical structures which always increases and that some compare to artistic works by M.C. Escher or J.S. Bach ...

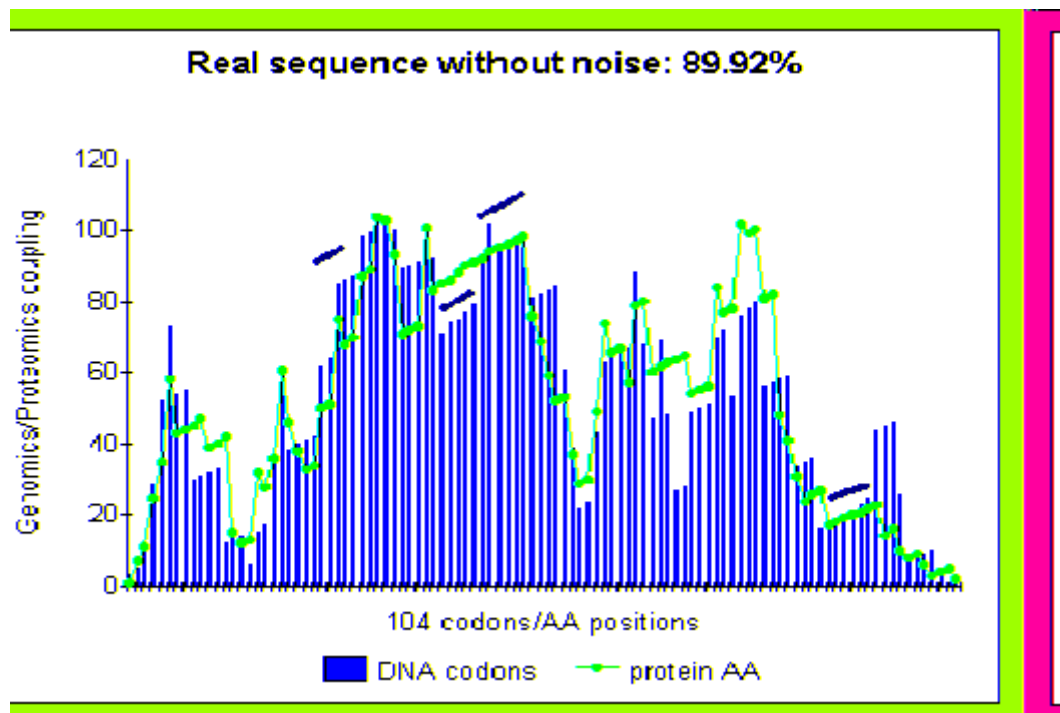


Figure 6 – "J.C. Perez's Master Code" typical derived increasing slopes of the proteomics patterned signature.

In two images below we generalized this study to whole human chromosome 22 studied in its entirety. By convention, the sequence is segmented into small pieces (eg 10000 bases pairs) and the total percentage textures "increase" type (growing) are recorded. The structures that we will discover are fractal in nature because they keep self-similar for smaller segmentations 1000 base pairs, or larger than 100000, 1000000 bases pairs.

If this work is carried out for genomic and proteomic picture, we see that if this trend seems self organized around one attractor for DNA double strand (genomics), it shows 2 levels, two "attractors" for the second (proteomics). A curious fact emerges then: although two curves genomics and proteomics are still highly correlated in their respective forms and shapes, we will discover that their textures are radically different, Thus, the population of genomic curves will be relatively dispersed around ONE single withdrawing attractor in a kind of Gaussian dispersion, while the population of proteomic curves be distributed around two attractors, bringing out a kind of binary frequency modulation.

We are witnessing "the emergence" of the "birth" of a Binary Code !

Let us not forget that the initial information was the atomic mass of each bio-atom, which is... a "real" (decimal) number!

Then, it is transformed into a code which is an integer number...

And it now emerges Binary Code, then 0/1 bits binary numbers!

Preliminary analysis shows that the average levels of these two attractors are around 0.61 (61%) and 0.30 (30%) then appear to be in a ratio of 2 ... We will return to these two values bellow...

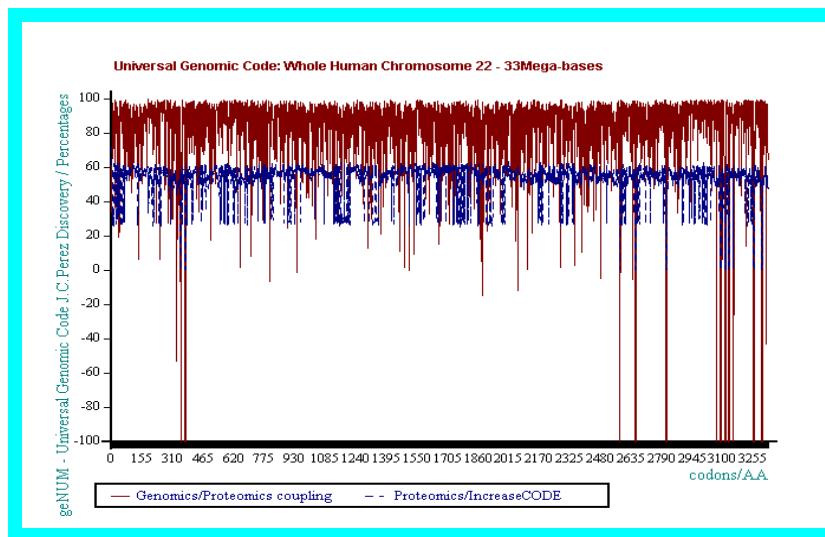


Figure 7 – Human chromosome 22 Proteomics Binary code (blue) and Genomics Analogic code (brown).

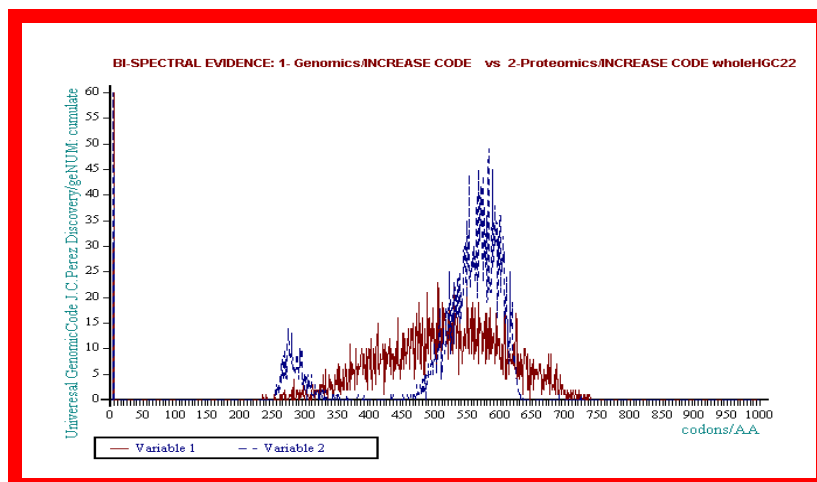


Figure 8 – Human chromosome 22 Attractors evidence: Proteomics Binary code (blue) and Genomics Analogic code (brown).

The figure bellow shows perfection of these two Proteomics attractors signature of one of the largest human chromosomes, chromosome 4 ...

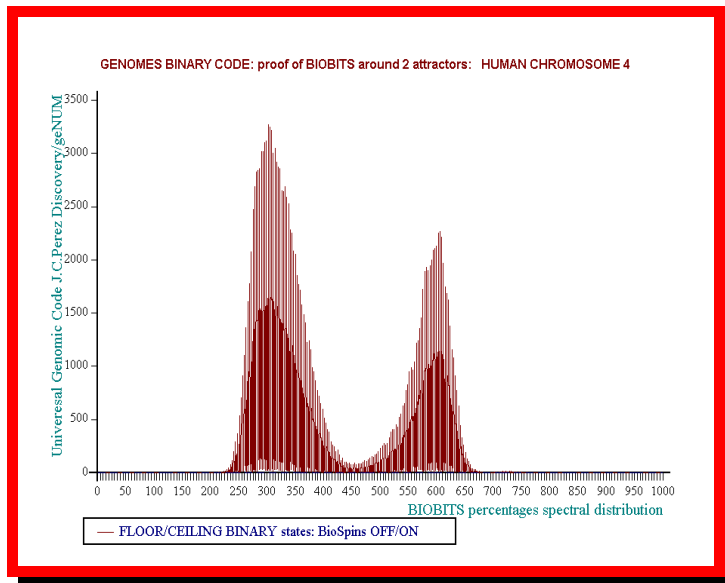


Figure 9 – Human chromosome 4 Binary Attractors evidence of the Proteomics Binary code texture signature.

DISCRETE WAVEFORMS: The emergence of “a modulated waveforms code” from the GENOMICS Master Code of any DNA sequence:

The generalization of previous gradients differentiations to what would appoint mathematics second, third or nth gradients differentiation orders will now highlight either "bits"... But "waveforms", more precisely discrete waveforms which we will measure periods: period of short-wave or 2 or 3 or even medium-wave wavelengths (greater than 10 times) ...

Thus, we calculate exhaustively all successive gradients or slopes: $S(i, i + 1)$, and $S(i, i + 2)$, $S(i, i + 3)$, ... $S(i, i + n)$. From all these successive gradients periodicities will then emerge.

Shortwave period=4 codons then 12 bases pairs in one million bases pairs within human chromosome 3 ...

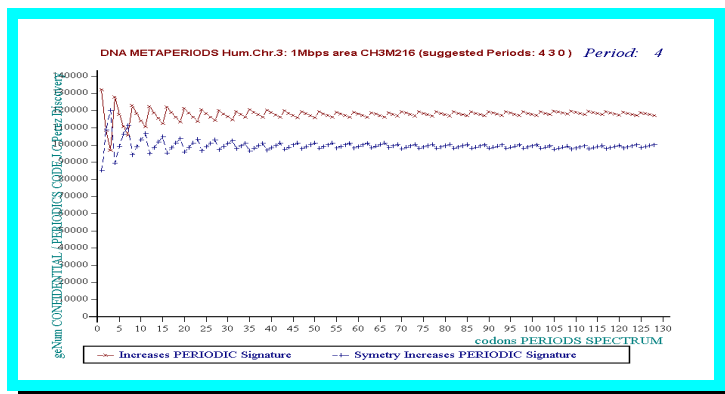


Figure 10 – Evidence of discrete waveforms within 1 million bases pairs in human chromosome3.

Long wave period=12 codons then 36 bases pairs in the first 300000 bases pairs within one of the largest human genes, the genetic disease Duchenne DMD gene.

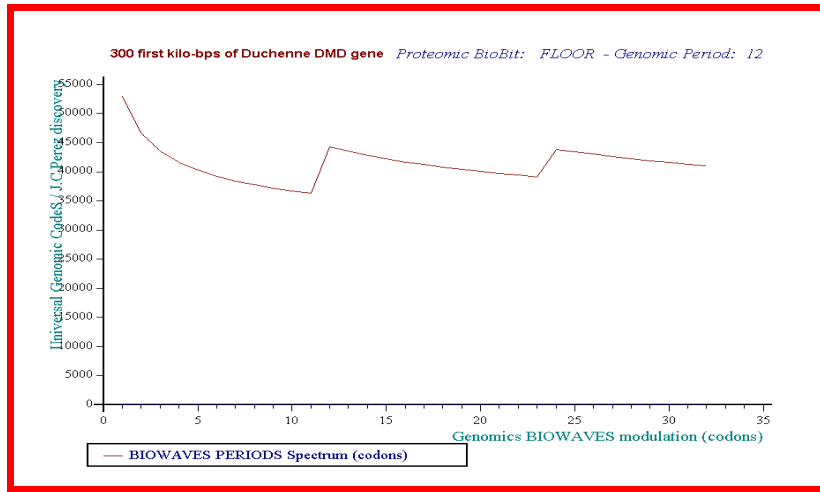


Figure 11 – Evidence of discrete waveforms within 300k-bases pairs from the DMD long Duchenne disease gene.

4. Applications

4.1 Functionality of Master Code Waveforms: the explanation of chromosomal alternated grey interferences bands:

One of the experimental concrete chromosomes representation is universally known Karyotypes image ...

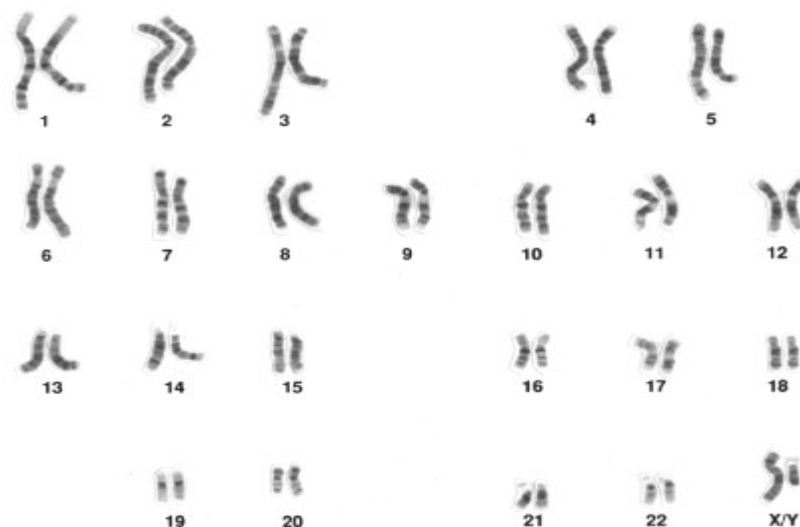


Figure 12 – The Universal Human Genome chromosomes banding called Karyotypes.

However, the synthesis of two earlier codes (binary code and waveforms) allows predict throughout the whole human genome, the alternating black / gray / white bands of Karyotypes! This is the clearest proof of the Functional reality of our Master Code discovery. We must recall here that karyotypes are obtained by interferometry, physical process of wave nature. The following is an illustration of the "calibration" in a portion of human chromosome 8 ...

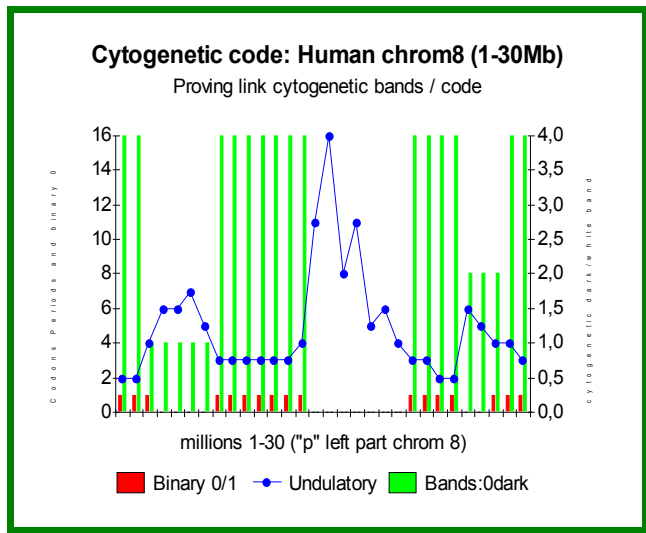


Figure 13 – The Master Code Biobits and discrete waveforms provide a perfect prediction of Karyotypes bands.

In Figure 13 above, the green bars represent the actual referenced colors of the karyotypes bands for this region of human chromosome8: 0 = black, 1 = dark gray, 2 = medium gray, light gray = 3, and 4 = white. Blue curve, modulation of waves predicted by the textures of the Master code: there is a fairly good correlation with the actual referenced colors of karyotypes, particularly, low wave period (2 or 3) for karyotypes clear (white), and waves heavy periods (in this case up to 16) for dark karyotypes (black). Finally, Biobits, shown here in red, have a status = 1 = "increase" for karyotypes clear (white) and a status = 0 = "decrease" for dark karyotypes (black).

Such predictive analyzes were performed for 24 chromosomes representing the entire human genome. The results show a perfect correlation between the predictions from the textures of the Master Code and grayscale all the karyotypes as they have been highlighted by the global community of geneticists [6]. Both Figures 14 and 15 below show a graphical summary modulations textures (Genomics) and Biobits (proteomics) throughout the whole human genome.

Notes:

-In each graph, the base unit analyzed X (horizontally) is one million base pairs long: then here 3266 units representing 3,266,000,000 bases pairs. Among them, 3.075 billion bases are significant, whereas the remaining 191 000 000 relate to GAPs (bases "N" indeterminate bases pairs), in particular centromeres regions of the chromosome.

-Vertical lines delimit the boundaries between each chromosome as centromere regions.

-The two variations shown here correspond to DNA Master Code textures (Genomic) and amino acid like Master Code textures (Proteomics). They are computed independently for each of the millions of bases analyzed, or "one point" per million bases on the Genomics curve and "one point" per million bases on the Proteomics curve.

-The reader will note again here that, although the two curves are variations Genomics and Proteomics strongly-correlated (96.63% on average over the whole genome), their "textures" are radically different! This result is quite remarkable showing both the fractal nature of the genome (roughness), and certainly, an undulatory waves based nature of any genomic DNA, the reality and the evidence will be recognized, probably within a few years (Montagnier 2009).

The first 8 chromosomes (1 to8) of whole HUMAN GENOME ...

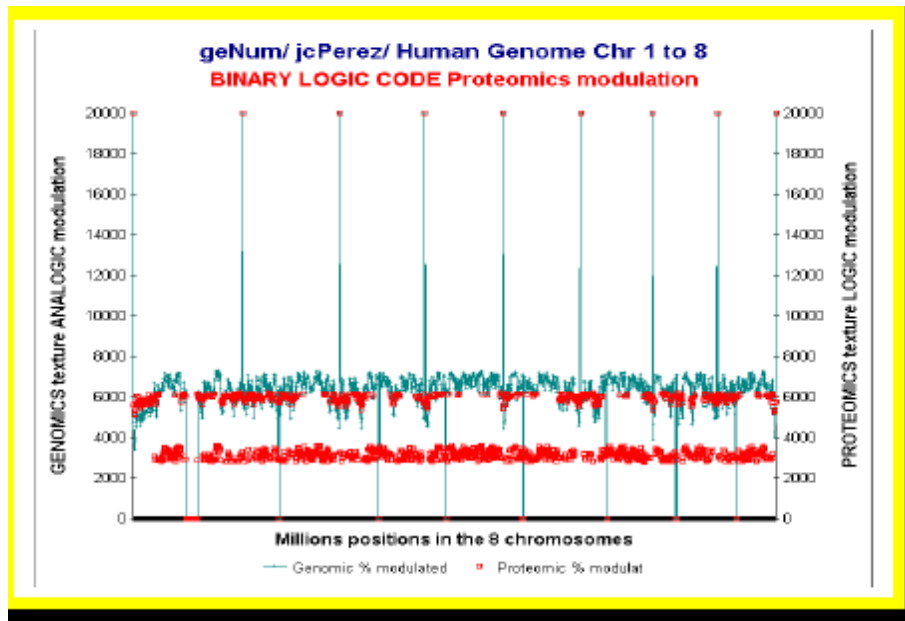


Figure 14 – Evidence of analogic modulated Genomics code (blue) and logic binary Proteomics code in the cases of human chromosomes 1 to 8.

The 16 last Chromosomes (9 toY) of the whole HUMAN GENOME ...

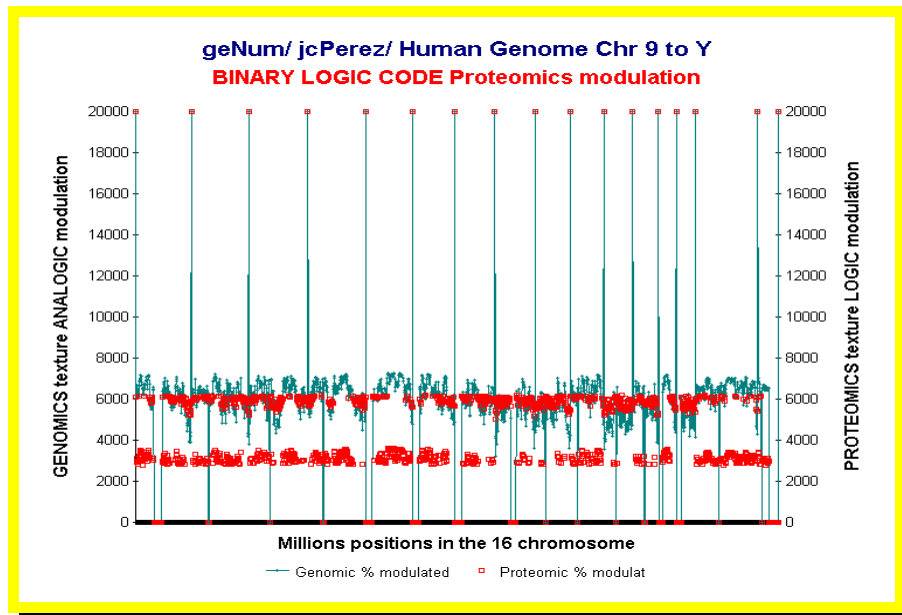


Figure 15 – Evidence of analogic modulated Genomics code (blue) and logic binary Proteomics code in the cases of human chromosomes 9 to Y.

In fact, the GENOMICS texture is “ANALOGICALLY MODULATED” around a region of about 60% mean value (6000 scale) would appear to be ... $F_i = 1/\Phi = 0.618$.

On the contrary, the PROTEOMICS texture (although strictly calculated using the same method and from strongly-correlated Master Code curves) is "MODULATED IN A BINARY LOGIC" continuously oscillating between two attractors whose respective values are: 30% = FLOOR average ceiling or 60% = CEILING average. The relationship between these two

attractors is very close to the number "TWO". The "clouds" of points perfectly illustrate the reality of these two binary attractors 0 / 1 or "FLOOR / CEILING".

In fact, an analysis of these average values over the entire human genome (Perez 2009) will reveal this remarkable result:

Average CEILING values % = 0.618 = 1 / Phi

while average FLOOR values % = 0.309 = 1/2 Phi. So CEILING / FLOOR ratio = 2.

4.2. An example of strong application of the Master Code: SNPs:

The many potential applications of Master Code of Biology were discussed briefly in a few conferences and publications: principally, (Perez 2000; Perez 2001) and informal WEB published drafts (Perez 2008; Perez 2001) describing the unification of the 3 languages of biology: DNA, RNA and amino acids chains, here we show that the SNPs are more important and functional by their location within the genomic DNA sequence than by their nucleotides mutations! As illustrated by fractal embedded russian dolls like figures 16 to 20... In this context of synchronicity between 2 Genomics Proteomics Master Code images, "peaks" and "valleys" will suddenly make a functional significance. This is the case in the following 5 successive zooms around a specific SNP in the human genome (SNP AL391359 from chromosome 21). This strange property of SNPs was tested on large populations SNP: it shows that the SNP, which have known that they distinguish each individual vis-à-vis all other humans, are most important in their states in the POSITION genome by the value of the CHANGE (exp: T ==> A). Here are 5 successive zooms nested in the same region of DNA located either side of the SNP (vertical line). They will highlight the fact that the SNP was "nestled" exactly in a "hollow". In other cases, it will nest on top instead of a "peak" ... The reader can check the universality of this SNPs property in extending the study to other positions of SNPs (which we did exhaustively).

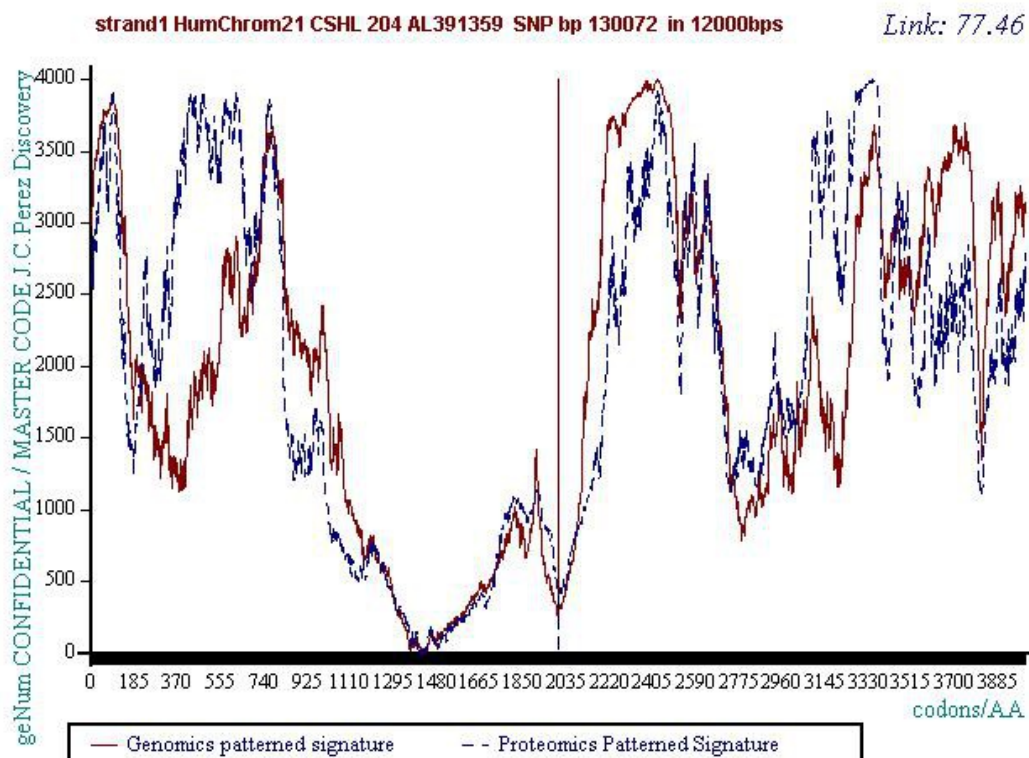


Figure 16 – “J.C. Perez’s MASTER CODE” Evidence of the functional role of SNP location within the chromosomes DNA sequences: here the 12000bp sequence around SNP.

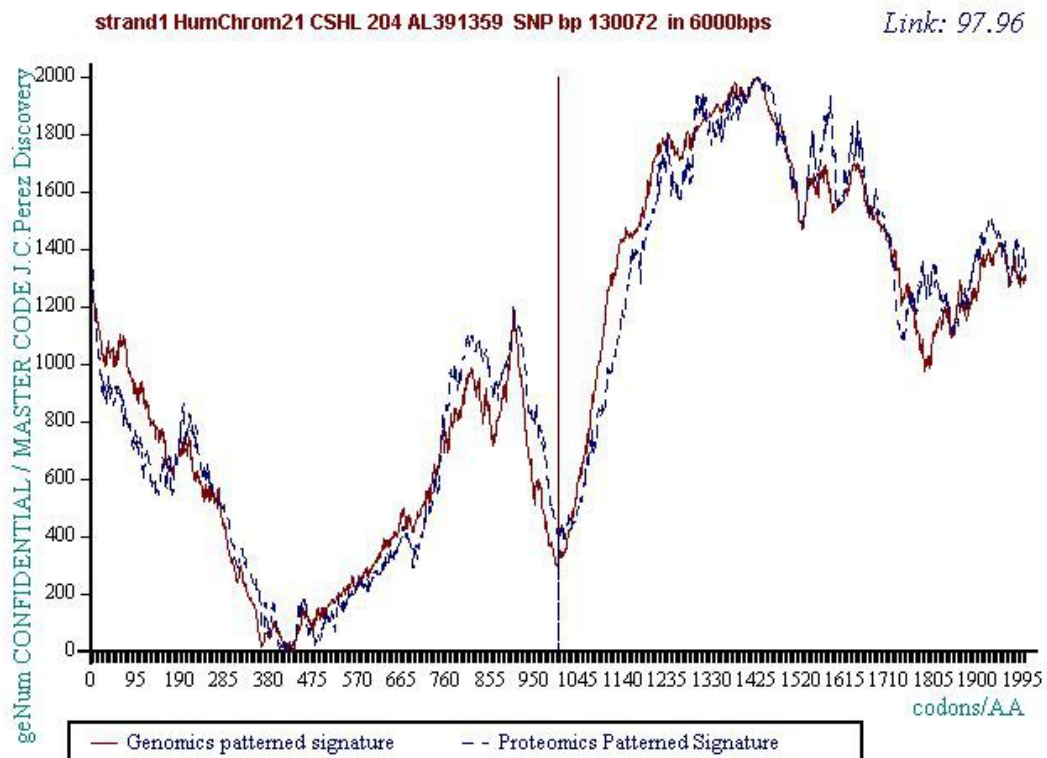


Figure 17 – “J.C. Perez’s MASTER CODE” Evidence of the functional role of SNP location within the chromosomes DNA sequences: here the 6000bp sequence around SNP.

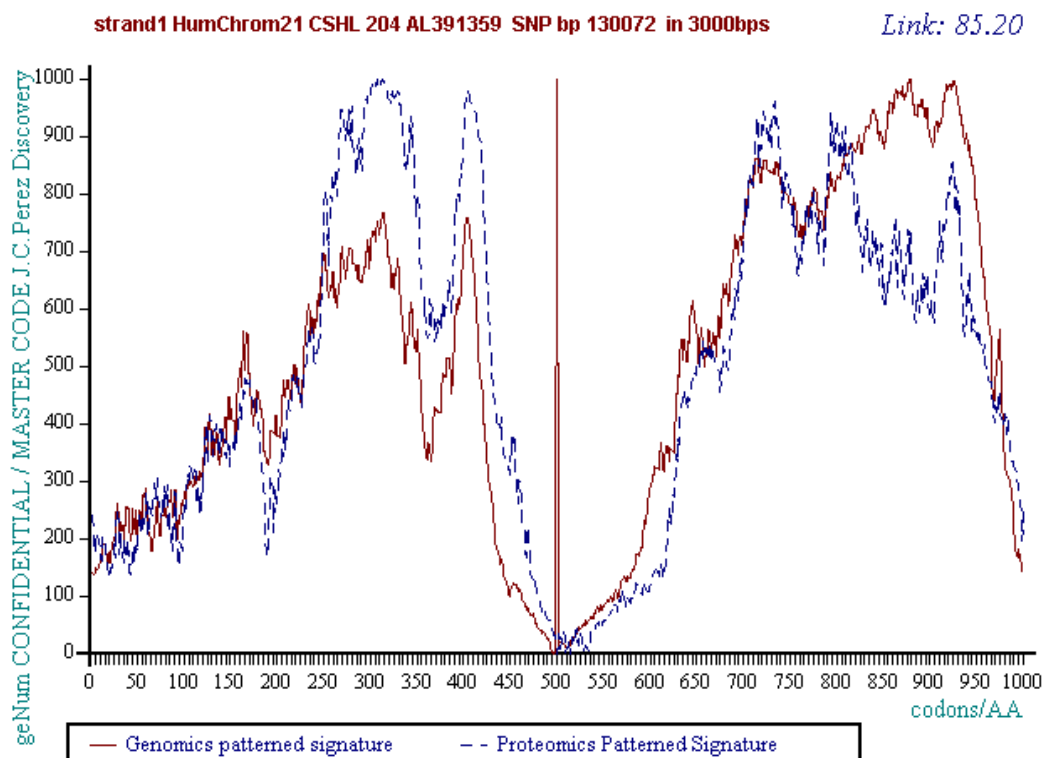


Figure 18 – “J.C. Perez’s MASTER CODE” Evidence of the functional role of SNP location within the chromosomes DNA sequences: here the 3000bp sequence around SNP.

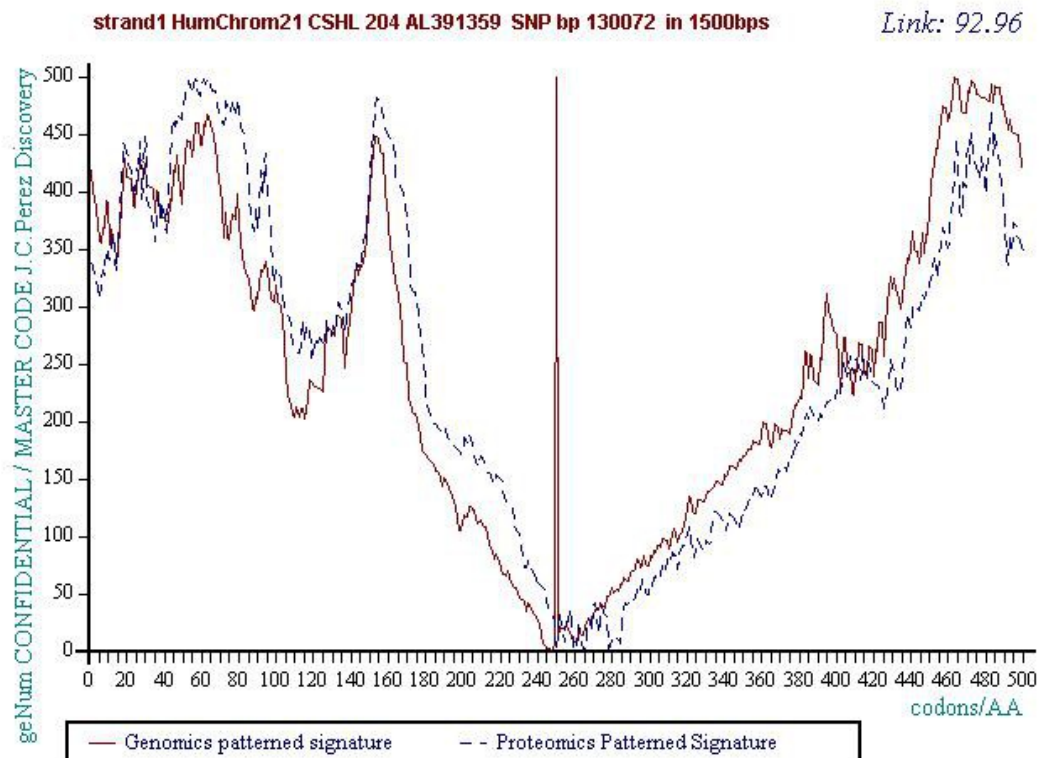


Figure 19 – “J.C. Perez’s MASTER CODE” Evidence of the functional role of SNP location within the chromosomes DNA sequences: here the 1500bp sequence around SNP.

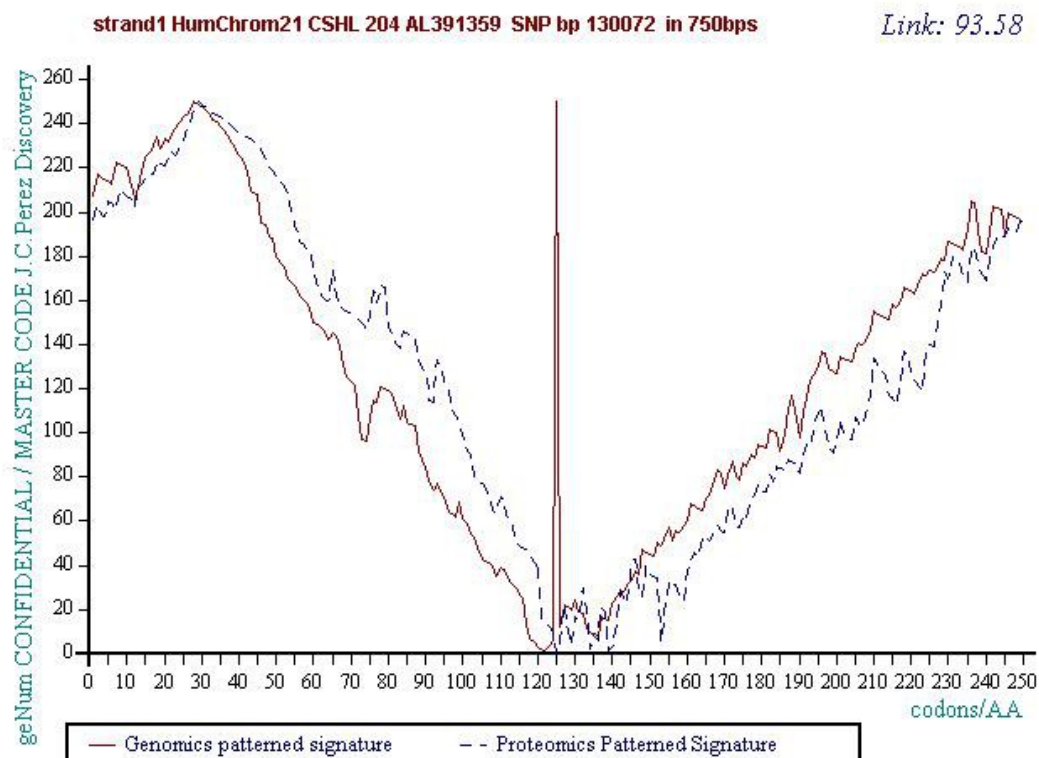


Figure 20 – “J.C. Perez’s MASTER CODE” Evidence of the functional role of SNP location within the chromosomes DNA sequences: here the 750bp sequence around SNP.

5. Conclusions

Since 2001 and the achievement of the Human Genome Project (Baltimore 2001), the whole human genome sequence is available and public. But what about ?

In one hand, the fractal nature of DNA and human genome is now a basic evidence (Pellionisz et al 2012) at global physical structure level (Lieberman-Aiden et al. 2009), and at codon populations level (Perez 2010).

Secondly, in our 25 years long research on numbers based patterns in DNA sequences (Marcer 1992), we evidenced multiples structures based on Fibonacci numbers and Golden Ratio numerical attractor. Today, others researchers demonstrate also various genetic areas controlled by golden ratio, like, here, RNA folding (Li. Hengwu et al. 2014).

Probably this *Master Code of Biology* discovery is a central key to understand Epigenetics particularly then finally discover the shaded function of Junk-DNA. Here we demonstrate only its crucial function in SNPs and at whole human genome structure scale.

Now, as described in §3, we proof that ***nucleotides sequences based waveforms, dissonances and resonances (Perez 2015b) are structuring and overlapping DNA sequences towards whole chromosomes***. These numerical waves recall a possible relationship with Nobel prizewinner Luc Montagnier strong results published in « ***DNA, waves and water*** » main article (Montagnier et al. 2009).

Theoretically, a remarkable result will deepen how unique information, the sequence of double-stranded genomic DNA, will "split" by 2 languages (DNA and amino acids) to produce two patterned images that are always highly correlated (more than 96% for the whole human genome)?

It is therefore surprising that the respective textures of each of these two curves differ strongly correlated so as contrasted: a diffusion around a single attractor for the Genomics image, recalling a kind of electronics AMPLITUDE MODULATION. Instead, the Proteomics image self-organizes around two attractors forming a sort of binary code swapping recalling electronics FREQUENCY MODULATION.

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